

across a vast array of skills. By using GPT-4, we were able to efficiently label a wide range of skills, allowing for a comprehensive and consistent categorization of all job postings in the online labor market. We minimized errors using state-of-the-art prompt engineering, measuring accuracy with a test set, and performing manual checks.

In addition, this paper focuses on the partial equilibrium effects of new technology and does not capture potential productivity increases outside of the platform under study. Technological improvements can decrease worker demand, but productivity gains can lead to new capital and organizational investments, potentially increasing labor demand (Acemoglu and Restrepo (2018), Autor and Salomons (2018)). To better understand how substitution of freelancers with GenAI tools affects in house labor, one would need to model the fragmentation of production into tasks in more detail (Rubbo, 2023).

A further limitation is that we only observe labor demand but not wages. Thus, we cannot definitively attribute changes in job postings to shifts in the demand curve. Workers might decrease their wage bids if they face more competition, or they might increase their wage bids if they believe ChatGPT enhances their value. However, previous research (Dube et al., 2020; Duch-Brown et al., 2022; Horton, 2021) suggests that online labor market workers have low bargaining power and act as price-takers. Moreover, our analysis indicates, that our results are not substantially biased by simultaneity issues arising from labor supply responses to the ChatGPT launch. Specifically, we demonstrate that while the launch of ChatGPT led to a decrease in labor demand, average employer budgets remained unchanged. Additionally, we observe an increase in the number of applicants per job posting is of the same magnitude than the reduction in job postings. These findings collectively support the conclusion that ChatGPT's launch led to reduced labor demand, with workers competing for a smaller pool of substitutable jobs.

While this analysis focuses on the short-term impacts of LLMs, understanding the long-term effects requires examining how these impacts vary based on workers' skills (Teutloff et al., 2023; Mealy et al., 2018), location (Farinha et al., 2020; Braesemann et al., 2022), and the alignment between current skills and those needed for emerging jobs (Neffke et al., 2024; Neffke and Henning, 2013). Our results provide evidence of first-order demand changes, which can be integrated into more complex models of worker adaptation (del Rio-Chanona et al., 2021; Acemoglu and Autor, 2011) to answer the broader question on the long-term effects of LLMs. Future research could examine the timing of demand shifts more precisely, investigating how long it takes employers to incorporate AI-driven tools and what the transition period looks like. Looking more at the firm side could reveal more about how to generalize from findings on contract work platforms to traditional labor markets. In addition, further work could explore whether the tasks whose demand decreased are now performed more effectively, performed differently, or not performed at all.

Our study reveals that a greater proportion of skill clusters are complemented by AI rather than substituted. However, while some of these complementary skill clusters experienced an increase in demand, others saw a decline. This underscores the critical role of public and private decision-makers in shaping the socio-economic impact of AI (Ritala et al., 2023; Gerling et al., 2024). Managers and policy-makers need to support workers in adapting and reskilling and should empower them to use LLMs in complementary ways (Retkowsky et al., 2024; Mollick et al., 2024). Our findings from online labor markets, which adapt quickly to technological changes, provide insights into potential future effects on traditional labor markets. Although traditional markets may adapt more slowly due to contract and severance laws, they will likely follow similar patterns over time. Policymakers must implement reskilling and upskilling programs to help workers transition into roles that complement AI technologies, focusing on foundational skills (Hosseinioun et al., 2023) and analytical thinking (Humburg and Van der Velden, 2017). Overall, our study demonstrates that while AI tools like ChatGPT can substitute some jobs, it also drives the development of new products and services, exemplifying creative destruction (Klimek et al., 2012) and contributing to the economy's evolution towards greater complexity (Balland et al., 2022).

Statements

Declaration of generative AI in scientific writing

During the preparation of this work the authors used ChatGPT 4.0 in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Funding information

Otto Kässi's work was funded by TT-säätiö (project title: "Tekoäly työelämässä – Miten käy Suomen kilpailukyvyyn?").

R. Maria del Rio-Chanona's research was supported by the James S. McDonnell Foundation.

Fabian Braesemann is supported by funding from the Oxford Internet Institute's Research Programme on AI & Work, funded by the Dieter Schwarz Stiftung gGmbH.

We are grateful for the support obtained by Open AI in providing access to the GPT API.

The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Online appendix: supplementary analyses

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jebo.2024.106845>.

Data availability

The authors do not have permission to share data.

References

- Acemoglu, D., Autor, D., 2011. Skills, tasks and technologies: Implications for employment and earnings. In: *Handbook of Labor Economics*, vol. 4, Elsevier, pp. 1043–1171.
- Acemoglu, D., Restrepo, P., 2018. The race between man and machine: Implications of technology for growth, factor shares, and employment. *Am. Econ. Rev.* 108 (6), 1488–1542.
- Adams-Prassl, A., Balgova, M., Qian, M., 2020. Flexible work arrangements in low wage jobs: Evidence from job vacancy data.
- Aggarwal, C.C., Hinneburg, A., Keim, D.A., 2001. On the surprising behavior of distance metrics in high dimensional space. In: *Database Theory—ICDT 2001: 8th International Conference London, UK, January 4–6, 2001 Proceedings* 8. Springer, pp. 420–434.
- Albanesi, S., da Silva, A.D., Jimeno, J.F., Lamo, A., Wabitsch, A., 2023. New Technologies and Jobs in Europe. Technical Report, National Bureau of Economic Research.
- Anderson, K.A., 2017. Skill networks and measures of complex human capital. *Proc. Natl. Acad. Sci.* 114 (48), 12720–12724.
- Arntz, M., Gregory, T., Zierahn, U., 2016. The Risk of Automation for Jobs in OECD Countries: A Comparative Analysis. oecd.
- Arthur, W.B., 2010. *The Nature of Technology: What it is and How It Evolves*. Penguin UK.
- Ash, E., Hansen, S., 2023. Text algorithms in economics. *Annu. Rev. Econ.* (ISSN: 1941-1391) 15 (1), 659–688. <http://dx.doi.org/10.1146/annurev-economics-082222-074352>.
- Autor, D., 2024. Applying AI to Rebuild Middle Class Jobs. Technical Report, National Bureau of Economic Research.
- Autor, D., Chin, C., Salomons, A., Seegmiller, B., 2024. New frontiers: The origins and content of new work, 1940–2018. *Q. J. Econ.* qjae008.
- Autor, D., Salomons, A., 2018. Is Automation Labor-Displacing? Productivity Growth, Employment, and the Labor Share. Technical Report, National Bureau of Economic Research.
- Balland, P.-A., Broekel, T., Diodato, D., Giuliani, E., Hausmann, R., O'Clery, N., Rigby, D., 2022. The new paradigm of economic complexity. *Res. Policy* 51 (3), 104450.
- Barnichon, R., 2010. Building a composite help-wanted index. *Econom. Lett.* 109 (3), 175–178.
- Bhattacharjee, P., Mitra, P., 2021. A survey of density based clustering algorithms. *Front. Comput. Sci.* 15, 1–27.
- Braesemann, F., Stephany, F., Teutloff, O., Kässi, O., Graham, M., Lehdonvirta, V., 2022. The global polarisation of remote work. *Plos one* 17 (10), e0274630.
- Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D.M., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., Amodei, D., 2020. Language models are few-shot learners. *CoRR*, abs/2005.14165. URL: <https://arxiv.org/abs/2005.14165>.
- Brynjolfsson, E., Li, D., Raymond, L.R., 2023. Generative AI at Work. Technical Report, National Bureau of Economic Research.
- Brynjolfsson, E., McAfee, A., 2014. *The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies*. WW Norton & Company.
- Brynjolfsson, E., Mitchell, T., 2017. What can machine learning do? Workforce implications. *Science* 358 (6370), 1530–1534.
- Corporaal, G.F., Lehdonvirta, V., 2017. Platform Sourcing: How Fortune 500 Firms are Adopting Online Freelancing Platforms. University of Oxford.
- Davis, S.J., Faberman, R.J., Haltiwanger, J.C., 2013. The establishment-level behavior of vacancies and hiring. *Q. J. Econ.* 128 (2), 581–622.
- del Rio-Chanona, R.M., Hermida-Carrillo, A., Sepahpour-Fard, M., Sun, L., Topinkova, R., Nedelkoska, L., 2023. Mental health concerns precede quits: shifts in the work discourse during the Covid-19 pandemic and great resignation. *EPJ Data Sci.* 12 (1), 49.
- del Rio-Chanona, R.M., Laurentsyeve, N., Wachs, J., 2024. Large language models reduce public knowledge sharing on online Q&A platforms. *PNAS nexus* 3 (9), pgae400.
- del Rio-Chanona, R.M., Mealy, P., Beguerisse-Díaz, M., Lafond, F., Farmer, J.D., 2021. Occupational mobility and automation: a data-driven network model. *J. R. Soc. Interface* 18 (174), 20200898.
- Demirci, O., Hannane, J., Zhu, X., 2023. Who is AI replacing? The impact of ChatGPT on online freelancing platforms. *The Impact of ChatGPT on Online Freelancing Platforms* (October 15, 2023).
- Dube, A., Jacobs, J., Naidu, S., Suri, S., 2020. Monopsony in online labor markets. *Am. Econ. Rev. Insights* 2 (1), 33–46.
- Duch-Brown, N., Gomez-Herrera, E., Mueller-Langer, F., Tolan, S., 2022. Market power and artificial intelligence work on online labour markets. *Res. Policy* 51 (3), 104446.
- Eloundou, T., Manning, S., Mishkin, P., Rock, D., 2023. Gpts are gpts: An early look at the labor market impact potential of large language models. *arXiv preprint arXiv:2303.10130*.
- Farinha, T., et al., 2020. Impacts From Automation Diffuse Locally – A Novel Approach To Estimate Jobs Risk In Us Cities. Technical Report, Utrecht University, Department of Human Geography and Spatial Planning
- Frey, C.B., Osborne, M.A., 2017. The future of employment: How susceptible are jobs to computerisation? *Technol. Forecast. Soc. Change* 114, 254–280.
- Gerling, C., Teubner, T., Braesemann, F., 2024. Who uses generative AI, how, and why? A cluster analysis on motives, perceptions, and use patterns of ChatGPT. *Electronic Markets* (In revision).
- Grootendorst, M., 2022. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. *arXiv preprint arXiv:2203.05794*.
- Grootendorst, M.P., 2024a. Outlier reduction - BERTopic — maartengr.github.io. https://maartengr.github.io/BERTopic/getting_started/outlier_reduction/outlier_reduction.html. (Accessed 20 May 2024).
- Grootendorst, M.P., 2024b. Parameter tuning - bertopic — maartengr.github.io. https://maartengr.github.io/BERTopic/getting_started/parameter%20tuning/parametertuning.html#top_n_words. (Accessed 20 May 2024).
- Hansen, S., Lambert, P.J., Bloom, N., Davis, S., Sadun, R., Taska, B., 2023. Remote Work across Jobs, Companies, and Space. National Bureau of Economic Research, <http://dx.doi.org/10.3386/w31007>.
- Hoberg, G., Phillips, G., 2016. Text-based network industries and endogenous product differentiation. *J. Polit. Econ.* (ISSN: 1537-534X) 124 (5), 1423–1465. <http://dx.doi.org/10.1086/688176>.
- Horton, J.J., 2021. The Ruble Collapse in an Online Marketplace: Some Lessons for Market Designers. Technical Report, National Bureau of Economic Research.
- Horton, J.J., Tambe, P., 2015. Labor economists get their microscope: Big data and labor market analysis. *Big data* 3 (3), 130–137.
- Hosseinoun, M., Neffke, F., Youn, H., et al., 2023. Nested skills in labor ecosystems: A hidden dimension of human capital. *arXiv preprint arXiv:2303.15629*.

- Hui, X., Reshef, O., Zhou, L., 2023. The short-term effects of generative artificial intelligence on employment: Evidence from an online labor market. Available at SSRN 4527336.
- Humburg, M., Van der Velden, R., 2017. What is expected of higher education graduates in the twenty-first century? In: *The Oxford Handbook of Skills and Training*. Oxford University Press US, p. 201.
- Karjane, A., 2023. Machine-assisted mixed methods: augmenting humanities and social sciences with artificial intelligence. *arXiv:2309.14379*.
- Kässi, O., Lehdonvirta, V., 2018. Online labour index: Measuring the online gig economy for policy and research. *Technol. Forecast. Soc. Change* 137, 241–248.
- Klimek, P., Hausmann, R., Thurner, S., 2012. Empirical confirmation of creative destruction from world trade data. *PLoS One* 7 (6), e38924.
- Li, Z., Shi, Y., Liu, Z., Yang, F., Liu, N., Du, M., 2024. Quantifying multilingual performance of large language models across languages. *arXiv preprint arXiv:2404.11553*.
- Liu, J., Xu, X., Li, Y., Tan, Y., 2023. "generate" the future of work through AI: Empirical evidence from online labor markets. *arXiv preprint arXiv:2308.05201*.
- Lukac, M., 2021. Two worlds of online labour markets: Exploring segmentation using finite mixture models and a network of skill co-occurrence.
- Lysyakov, M., Viswanathan, S., 2023. Threatened by AI: Analyzing users' responses to the introduction of AI in a crowd-sourcing platform. *Inf. Syst. Res.* 34 (3), 1191–1210.
- MacKinnon, J.G., Webb, M.D., 2020. Randomization inference for difference-in-differences with few treated clusters. *J. Econometrics* (ISSN: 0304-4076) 218 (2), 435–450. <https://dx.doi.org/10.1016/j.jeconom.2020.04.024>, URL: <https://www.sciencedirect.com/science/article/pii/S0304407620301445>.
- Marinescu, I., Wolthoff, R., 2020. Opening the black box of the matching function: The power of words. *J. Labor Econ.* 38 (2), 535–568.
- McInnes, L., Healy, J., 2017. Accelerated hierarchical density based clustering. In: *2017 IEEE International Conference on Data Mining Workshops. ICDMW, IEEE*, pp. 33–42.
- McInnes, L., Healy, J., Melville, J., 2018. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426*.
- Mealy, P., del Rio-Chanona, R.M., Farmer, J.D., 2018. What you do at work matters: new lenses on labour. *What You Do at Work Matters: New Lenses on Labour* (March 18, 2018).
- Mikolov, T., Chen, K., Corrado, G., Dean, J., 2013. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.
- Mollick, E., Mollick, L., Bach, N., Ciccarelli, L., Przystanski, B., Ravipinto, D., 2024. AI agents and education: Simulated practice at scale. *arXiv preprint arXiv:2407.12796*.
- Moussiades, L., Zografos, G., Papakostas, G., 2024. GPT-4 vs. GPT-3.5 As Coding Assistants. Research Square Platform LLC, <http://dx.doi.org/10.21203/rs.3.rs-3920214/v1>.
- Neffke, F., Henning, M., 2013. Skill relatedness and firm diversification. *Strateg. Manag. J.* 34 (3), 297–316.
- Neffke, F., Nedelkoska, L., Wiederhold, S., 2024. Skill mismatch and the costs of job displacement. *Res. Policy* 53 (2), 104933.
- Noy, S., Zhang, W., 2023. Experimental evidence on the productivity effects of generative artificial intelligence. Available at SSRN 4375283.
- OpenAI, 2023. Best practices for prompt engineering with openai API. <https://help.openai.com/en/articles/6654000-best-practices-for-prompt-engineering-with-openai-api>. (Accessed 12 February 2024).
- Peng, S., Kalliamvakou, E., Cihon, P., Demirel, M., 2023. The impact of ai on developer productivity: Evidence from github copilot. *arXiv preprint arXiv:2302.06590*.
- Pennington, J., Socher, R., Manning, C.D., 2014. Glove: Global vectors for word representation. In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing. EMNLP*, pp. 1532–1543.
- Qiao, D., Rui, H., Xiong, Q., 2023. AI and jobs: Has the inflection point arrived? Evidence from an online labor platform. *arXiv preprint arXiv:2312.04180*.
- Retkowsky, J., Hafermalz, E., Huysman, M., 2024. Managing a ChatGPT-empowered workforce: Understanding its affordances and side effects. *Bus. Horiz.*
- Ritala, P., Ruokonen, M., Ramaul, L., 2023. Transforming boundaries: how does ChatGPT change knowledge work? *J. Bus. Strategy* (ahead-of-print).
- Rubbo, E., 2023. Fragmentation of production and the wage distribution. Working Paper.
- Stephany, F., Kässi, O., Rani, U., Lehdonvirta, V., 2021. Online labour index 2020: New ways to measure the world's remote freelancing market. *Big Data Soc.* 8 (2), 20539517211043240.
- Stephany, F., Teutloff, O., 2024. What is the price of a skill? The value of complementarity. *Res. Policy* 53 (1), 104898.
- Teutloff, O., Stenzhorn, E., Kässi, O., 2023. Skills, job application behavior and the gender wage gap: Evidence from online freelancing. Available at SSRN.
- Van Der Maaten, L., Postma, E.O., van den Herik, H.J., et al., 2009. Dimensionality reduction: A comparative review. *J. Mach. Learn. Res.* 10 (66–71), 13.
- Webb, M., 2019. The impact of artificial intelligence on the labor market. Available at SSRN 3482150.
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Chi, E.H., Le, Q., Zhou, D., 2022. Chain of thought prompting elicits reasoning in large language models. *CoRR* abs/2201.11903. URL: <https://arxiv.org/abs/2201.11903>.